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Artificial Intelligence Chatbots as Relational Agents: A Systematic Review of Human–AI Chatbot Relationships

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ABSTRACT

This study presents a systematic review guided by the *human–AI chatbot Relationship (HAIR) Framework of Social Influence* to clarify the antecedents of human–AI chatbot relationships and their social influence outcomes. Following PRISMA guidelines, we searched six databases (ACM Digital Library, Web of Science, Scopus, IEEE, EBSCO, and PsycINFO) and identified 16,358 records, of which 68 papers (78 studies) met inclusion criteria. Findings reveal that trust and perceived social support frequently functioned as relational mediators through which chatbots exert social influence. Although chatbot design features were most commonly examined, message-level factors such as empathy and responsiveness showed extensive associations with relational outcomes but remain underexplored. The literature is dominated by short-term experimental designs, inconsistent conceptualizations of key constructs, and Western samples, limiting understanding of long-term and cross-cultural relationship development. This study clarifies the relational processes underlying AI chatbot influence and offers directions for designing more effective human–AI communication.

KEYWORDS

Artificial intelligence; chatbot; human–AI relationship; social influence; trust

1. Introduction

Artificial intelligence (AI) chatbots are conversational agents that have the ability to continuously and naturally converse, without a pre-set script (Rapp et al., 2021). A growing body of research highlights their potential for social influence across diverse domains, including support provision (Beattie & High, 2022); health behavior promotion (Oh et al., 2021), business (Mariani et al., 2023), and education (Guo & Wang, 2025; Kuhail et al., 2023; Wang et al., 2025). Despite the increasing use of chatbots across diverse social influence contexts, chatbots are often framed in primarily instrumental terms, as tools for delivering information and executing human commands. Communication research, however, emphasizes that influence unfolds within the context of personal relationships (Katz & Lazarsfeld, 1955), pointing to the need to consider chatbots not only as functional tools but also as relational partners.

Bickmore and Picard (2005) conceptualized computational programs with which people can form personal relationships as *relational agents*. They argued that a central step in leveraging these agents for social influence is supporting users in building such relationships. Advances in AI, machine learning, and natural language processing (Brandtzaeg & Følstad, 2017) have since made it possible for AI chatbots to engage in increasingly natural and humanlike conversations. These technological developments have expanded chatbots' roles beyond task completion, enhancing their relational capabilities and paving the way for more complex human–chatbot interactions. Recent research underscores the potential of human–AI chatbot relationships, revealing that humans can develop relationships with chatbots in a similar way to how they establish relationships with other humans, characterized by multiple

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relationship dimensions such as trust (Ng & Zhang, 2025), rapport (Brixey & Novick, 2019), social connectedness and closeness (Christoforakos et al., 2021; Lee et al., 2020), and companionship (Skjuve et al., 2021). In line with this perspective, recent work has begun to explore human–AI chatbot relationship building through both interpersonal concepts, such as trust, and pseudo-personal concepts, such as anthropomorphism (for reviews, see Ng & Zhang, 2025, Rapp et al., 2021). Prior reviews have primarily adopted an HCI or user-experience lens, emphasizing themes such as satisfaction, engagement, acceptance, emotional involvement, and perceived humanness (Rapp et al., 2021). Other syntheses have narrowed their focus to a single relational construct, most notably trust, examining how it is defined, operationalized, and predicted (Ng & Zhang, 2025). While these efforts provide valuable insights, they do not systematically examine how relational mechanisms function as mediating pathways linking chatbot characteristics to downstream social influence outcomes. Consequently, a relationship-centered, theory-driven synthesis remains needed.

Studies suggest that AI chatbots that are perceived as personal contacts can have unique advantages in persuasion tasks such as reducing psychological reactance to social influence through relational communication, as compared to other types of AI persuasion technology (e.g., contemplator AI technologies which help make decisions such as AI doctors). Central to these are similar attributional processes like in relational interaction between humans, which suggests that relational communication theories can be applied to understanding how human–AI chatbot relationships are built and the impact on social influence (Bickmore & Picard, 2005). In this study, we define relationship building outcomes as perceptual and behavioral outcomes within users indicating how well users and AI chatbots are connected. Bickmore and Picard (2005) idea of relational agents leads us to propose the *Human–AI Chatbot Relationship* (HAIR) *Framework of Social Influence*. Centering the role of relational communication, the HAIR framework highlights how communication may foster organic, personal relationships between humans and AI chatbots, thereby shaping user perceptions, emotions, and behaviors. While the idea of social influence through personal relationships with AI chatbots is promising, existing findings have yet to be systematically synthesized under such a framework. Here, we present a timely systematic review under the HAIR framework. Specifically, we review contributing factors to human–AI chatbot relationships (relational outcomes), as well as how human–AI chatbot relationships predict social influence outcomes. In reviewing the contributors to relational outcomes from a communication perspective, we review findings relevant to user and chatbot characteristics that may impact communication processes as well as findings of the features of communication messages exchanged.

1.2. Antecedents of human–AI chatbot relationships: User, chatbot, and message

Existing research on human–AI chatbot relationships has primarily focused on the characteristics of chatbot users and the chatbots themselves. In terms of user characteristics, prior studies have examined the impact of demographic traits (e.g., Ki et al., 2020), prior experiences with AI chatbots, attitudes toward AI (e.g., Gnewuch et al., 2022), personal disposition to trust technology (Prakash et al., 2023), user personality (e.g., neuroticism: Meng & Dai, 2021), and the motivations and goals behind interactions (e.g., Brandtzaeg & Følstad, 2017) on human–AI chatbot relationships. Regarding chatbot characteristics, research has explored whether factors such as displayed gender (Zogaj et al., 2023), the degree of humanness or anthropomorphism (Jiang et al., 2023; Li & Sung, 2021; Youn & Jin, 2021), communication competence (Meng et al., 2023), responsiveness (Jiang et al., 2022), and the type of relationship formed (Youn & Jin, 2021) affect relational perceptions and behaviors.

However, beyond user and chatbot characteristics, message characteristics represent a critical yet often overlooked factor in the development and maintenance of human–AI chatbot relationships. Communication, at its core, involves the exchange of messages (Lasswell, 1948). The predominant message-centered approach by Burleson (2010) contends that interpersonal communication research should focus on studying messages exchanged in interpersonal interactions to understand how relationships are built, maintain, and function. The focus on messages highlights the pivotal role of message production and interpretation within social interactions, shaping both relational dynamics and broader social behaviors, thus offering precise insights for the construction of more favorable social outcomes (Burleson, 2010). The characteristics of messages exchanged between human and AI chatbots can play

a pivotal role in relationship development with AI chatbots (reciprocity: Meng & Dai, 2021; contingency: Meng et al., 2023). However, research that centers on the influence of message characteristics on building human–AI chatbot relationships remains relatively sparse. Meng and Dai (2021) investigated how a chatbot's self-disclosure and provision of emotional support impact individuals' stress and worry. Their findings revealed that reciprocal self-disclosure reduced worry, and self-disclosure could alleviate stress, but only when accompanied by emotional support. Other investigations into message characteristics have delved into the emotionality of the messages (Jiang et al., 2023), support and empathy expressed in messages (Lee et al., 2022), message contingency (Meng et al., 2023), disfluency (Bührke et al., 2021), and the level of personalization of the messages (Kim & Hur, 2024). Taken together, in addition to reviewing findings of user and AI chatbot characteristics that may impact communication processes and outcomes, we also review findings about communication message characteristics—all three are contributing factors of human–AI chatbot relationships.

1.3. Human–AI chatbot relationships and social influence outcomes

Recent literature examined the associations between human–AI chatbot communication and subsequent perceptual and behavioral outcomes (Meng & Dai, 2021; Park et al., 2022). We delineate these outcomes into two overarching categories: primary (relational) and secondary (social influence) outcomes. Primary outcomes are those directly tied to the immediate human–AI chatbot relationship. Specific message exchanges such as self-disclosure or emotional support forms human users' perceptions of intimacy (Park, Chueng, & Lee, Lee et al., 2023), trust (Chen et al., 2025), relational satisfaction (Jiang et al., 2022), and emotional bonding (Lee et al., 2022) with the AI chatbot they interact as well as their level of engagement. Also, message characteristics have the potential to elicit behavioral reactions, including subsequent self-disclosure of personal details as well as the expression of a range of emotions. These perceptual (e.g., rapport) and behavioral (e.g., self-disclosure) outcomes are classified as relational outcomes. The quality of a human–AI chatbot relationship can vary from nearly non-existent to highly intense and interdependent.

Interpersonal scholars contend that relational communication can strongly shape social influence outcomes (Dillard & Knobloch, 2011). Extending this idea, it is important to consider how human–AI chatbot communication may enhance or limit such outcomes. Prior studies have shown that favorable relational outcomes from human–AI chatbot communication are associated with improved psychological processes, such as reattributions and reappraisals of stressors (Meng et al., 2023), lower perceived stress (Meng & Dai, 2021), better emotional experience (Crolic et al., 2022), and higher self-efficacy (Croes & Antheunis, 2021). Studies have also found that relational outcomes are linked to secondary behavioral outcomes, such as adherence to health guidelines (Park, Chung, & Lee, 2022), educational achievement (De Nieva et al., 2020), and purchasing behaviors (Croes & Antheunis, 2021).

1.4. Overview of the current study

Drawing upon the aforementioned literature, we propose the HAIR framework of social influence (see Figure 1), to guide our systematic review. Briefly summarized, we see human–AI chatbot relationship building as the bridge between antecedents (i.e., characteristics of users, AI chatbots, and the messages exchanged) and social influence outcomes. Our review focuses closely on relationship-related variables as key mechanisms in this process. Based on this framework, we pose the following research questions.

RQ1: What characteristics of (a) chatbot, (b) user, and (c) message may influence human–AI chatbot relationships?
RQ2: What are potential social influence outcomes of human–AI chatbot relationships?

2. Methods

This systematic review has been registered in Open Science Framework (OSF; https://osf.io/xrqtd/?view_only=fd2f2e65b0734372a4d1e7e7d32a1e75). The Preferred Reporting Items for Systematic

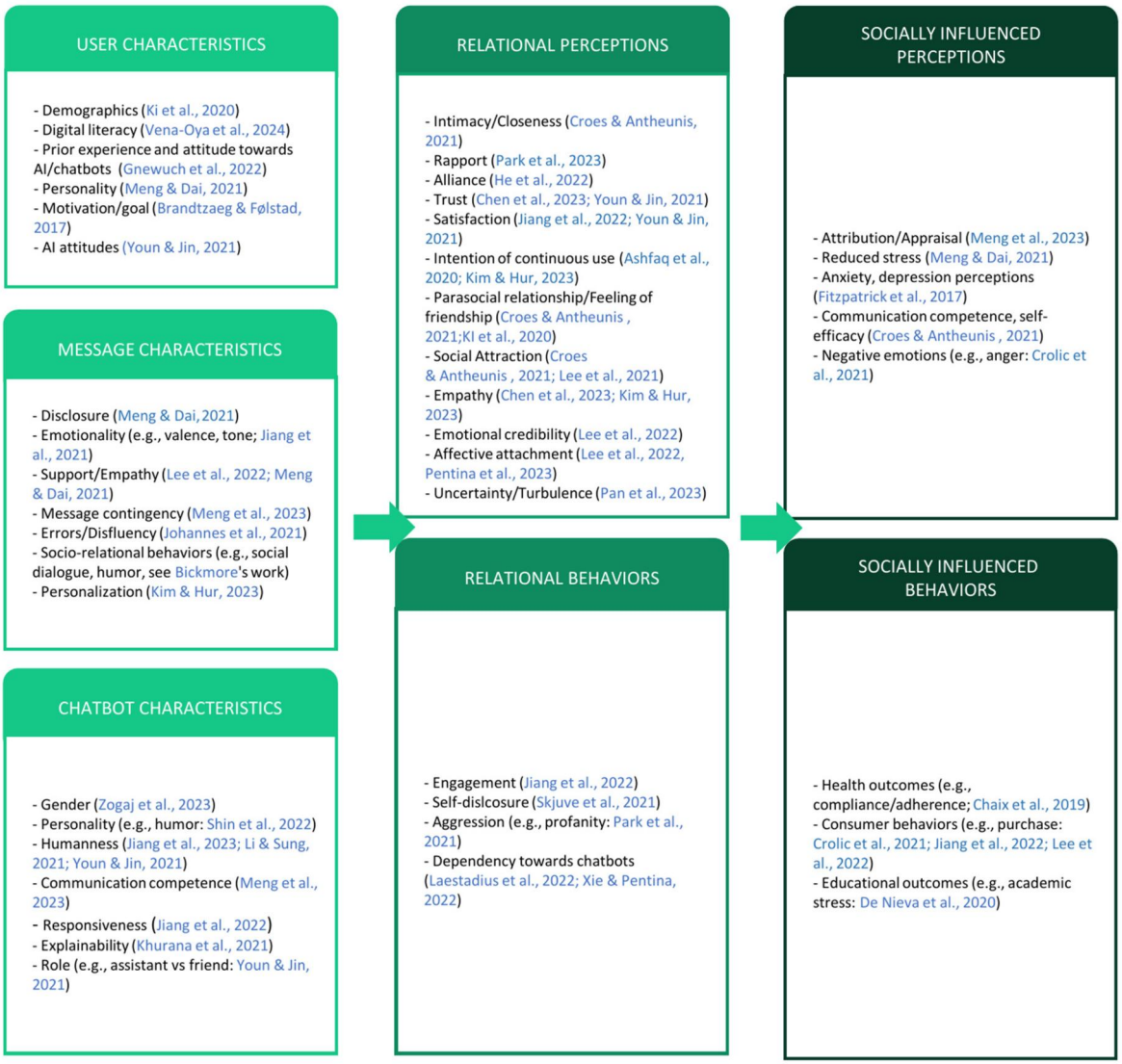


Figure 1. Human–AI relationship framework of social influence.

Reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021) has been used in this review (see Appendix A for the PRISMA 2020 checklist). This systematic review did not involve the collection of new data from human participants. Therefore, institutional ethical approval was not required.

2.1. Eligibility criteria

The eligibility of studies was determined by the following criteria: Study aims, outcomes, chatbot system feature, modality, study design, source type, publication date, and language (Table 1). For this review, we included studies on AI chatbots with explicit AI features, focusing on those that primarily deliver messages via text or speech, with emphasis on the human-AI interaction and its impacts. Eligible studies include cross-sectional, quasi-experimental, and randomized controlled trials, published in English in peer-reviewed journals or conference proceedings from January 1st, 2010, to December 31, 2023. We limited inclusion to 2010 onward to focus on modern AI-driven conversational agents capable of sustained and socially interactive communication, as earlier systems were primarily rule-based and less suitable for examining relational processes. This criterion aligned with prior reviews (e.g., Rapp et al., 2021).

Table 1. Inclusion and exclusion criteria.

	Inclusion criteria	Exclusion criteria
Study aim(s)	Studies that focus on the interaction between users and AI chatbot with particular attention to the human-side of such interaction.	Studies that assess the effectiveness of AI chatbots in performing specific, instrumental tasks, or their natural language processing technology or algorithms, without exploring the conversational & relational experience of the user with the chatbot.
Outcome(s)	Main outcomes: Perceptions and behaviors of human–AI chatbot relationships Secondary outcomes: Perceptual and behavioral outcomes relevant to health, education, and social influence	Studies that report only chatbot infrastructure, algorithm design, or performance outcomes (e.g., contingency)
Chatbot system feature	Chatbots under investigation have artificial intelligence (AI) features and the AI features are mentioned explicitly or elaborated.	Chatbots under investigation do not have AI features.
Modality	Chatbot conveys message through text or speech. In case or multi-modality interaction, either text or speech should be the main mode of interaction	Chatbot conveys messages mainly through visual cues (e.g., animated facial expressions)
Study design	Cross-sectional studies, quasi-experimental studies, or randomized controlled trials	Qualitative studies, case-control studies, reviews, descriptive studies, or cohort studies
Source type	Peer-reviewed journal articles or published conference proceedings	Papers published in the adjunct proceedings of conferences (e.g., posters, workshop papers, late-breaking result papers), reviews, books, or dissertations
Publication date	Published between January 2010 to December 2023	Published before 2010
Language	English	Non-English language

2.2. Information sources

A systematic search was conducted on six electronic interfaces, including ACM Digital Library, Web of Science Core Collection (WOS), SCOPUS, IEEE, EBSCO, and PsycINFO. References of the selected studies were checked to identify other eligible studies. Any additional potentially eligible studies were identified and screened. References were exported into the online software Covidence (www.covidence.org), and duplicates were removed using this tool.

2.3. Search strategy

The search query was a combination of chatbot-related keywords and relationship-related keywords. After reviewing past literature, the search terms were thoroughly examined and selected by the team. In each database, this search query was used to identify literature containing the search terms in the title or abstract. Detailed search strategies can be found in Appendix B.

2.4. Study selection

All retrieved references were reported into the Endnote reference management software (https://endnote.com), and duplicates were removed. The remaining references were imported into the Covidence systematic review software (http://www.covidence.org), and additional duplicates were removed. Before screening the articles, four researchers met to discuss the procedure for title and abstract screening using 20 randomly selected papers. Then, two researchers imported approximately 20% of total references for dual screening of titles and abstracts against the eligibility criteria. The agreement in the abstract and title screening between two reviewers was 96.7% (Krippendorff's alpha = 0.72). Then, the remaining abstract and title screening were done independently. Then, they both reviewed 25% of the full text. The agreement was 94.6% (Krippendorff's alpha = 0.86). After achieving a good level of inter-rater reliability, the remaining full texts were split into two and independently assessed. Discrepancies at each stage were resolved through discussion with a third reviewer.

2.5. Data collection process

Data extraction was conducted by two independent reviewers. Data collection forms were developed based on the Cochrane Handbook for Systematic Reviews of Interventions and the PRISMA strategy (Higgins et al., 2019). The data items to be extracted were pilot-tested using a Microsoft Excel spreadsheet and a sample of potential studies obtained through the initial iterative search process. A third reviewer was involved in resolving any disagreements and cross-checked the accuracy of the extracted data.

2.6. Data items

The data extraction form includes study characteristics (e.g., author, published year, study design, setting, and duration), participant characteristics (e.g., age, gender, education, sample size, race, and socioeconomic status), study design (e.g., survey or experiment), artificial intelligence under investigation (e.g., virtual companion, and chatbots), antecedents (e.g., characteristics of messages, AI chatbots, and users), secondary outcomes (e.g., health behavioral change), and safety and ethical assessment.

2.7. Data synthesis

We conducted a qualitative synthesis of findings from eligible studies, covering characteristics of AI chatbots, study designs, target population, and primary and secondary outcomes. Regarding risk of bias assessment, the Mixed Methods Appraisal Tool (MMAT; Hong et al., 2018) was used to evaluate the quality of the selected studies. Consistent with prior MMAT-based systematic reviews that summarize study rigor using percentage thresholds (e.g., Chu et al., 2020; El-Awaisi et al., 2018), studies meeting at least four or five criteria ($\geq$ 80%) were considered to be 'high quality'. We first selected 10 studies and all authors independently coded and cross-validated the coding through multiple rounds of discussion until we reached consensus. Then, we divided the rest of the studies to be assessed independently.

3. Results

3.1. Study selection

Figure 2 shows the study selection process. The search yielded 16,358 references in total, from which 8,631 duplicates were removed. A total of 7,727 abstracts were initially screened, among which 7,357 were deemed to be ineligible or manually identified as duplicates, leaving 370 to be read in full text. In total 68 papers (and 78 corresponding studies) have been included for review.

3.2. Risk of bias assessment

Seventy-eight studies were assessed using the MMAT (see Appendix C). We identified 48 quantitative randomized controlled trials, 2 quantitative non-randomized studies, and 27 quantitative descriptive studies. Among the studies, 8 randomized controlled trials, 1 non-randomized study, and 20 quantitative descriptive studies were regarded as high quality. Studies satisfying at least 80% (four or five) assessment criteria were considered as high quality. For quantitative randomized controlled trials, the criterion "Are the groups comparable at baseline?" was most frequently rated as unclear, with 83.3% of studies lacking sufficient reporting of baseline characteristics across groups. For quantitative descriptive studies, the most common limitation concerned sample representativeness, with 82.1% of studies rated as "No" or "Cannot tell" for whether the sample reflected the target population. These findings indicate that many studies did not adequately report baseline comparability or justify how samples represented their intended populations.

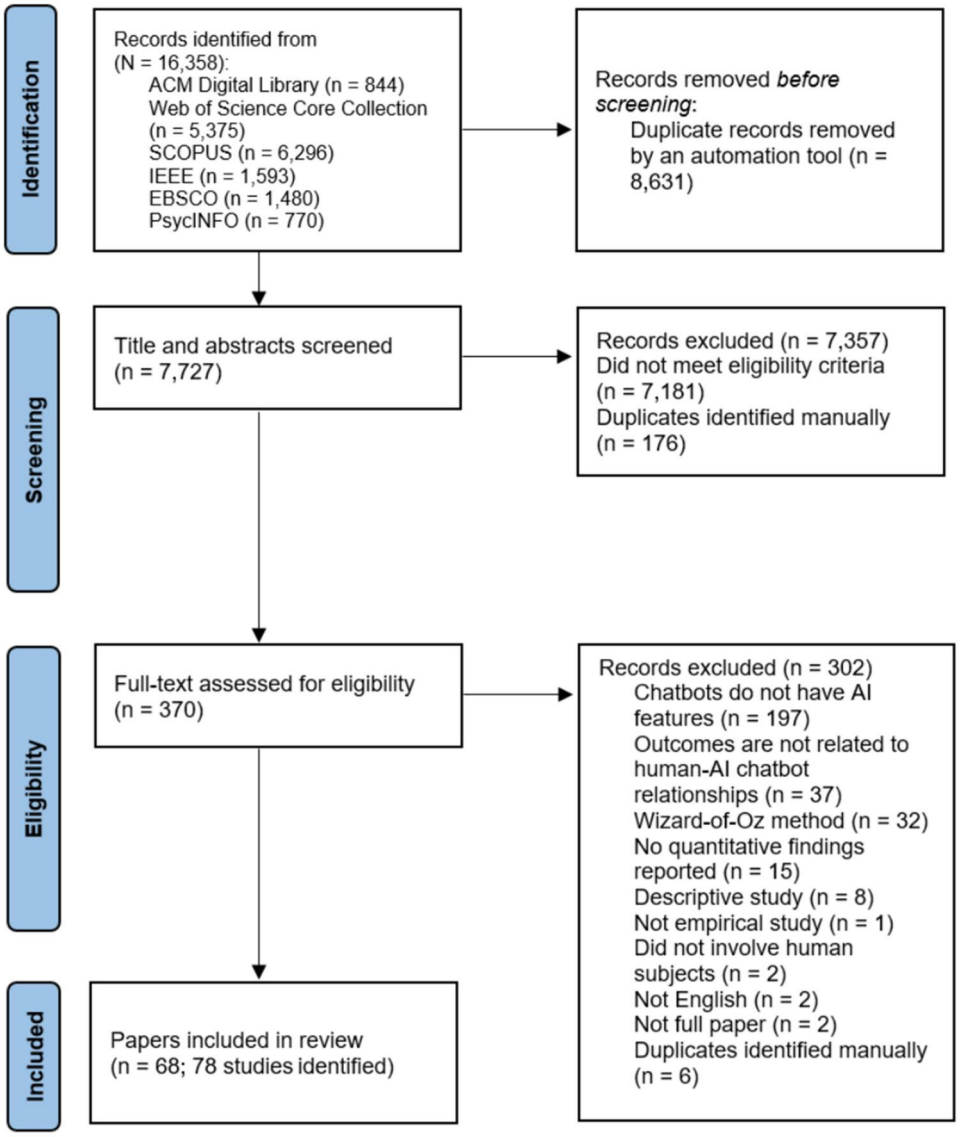


Figure 2. Flow diagram of the study selection process.

3.3. Descriptives of studies

Table 2 summarizes descriptives of studies. Among the 78 included studies, about two-thirds used experimental designs (primarily online), roughly one-third employed surveys, and only a handful were content analyses. They were predominantly published between 2021 and 2023, with sample sizes ranging from under 100 to over 500 participants. Studies commonly focused on specific groups (e.g., students or particular language/experience). Geographically, most studies were conducted in Europe or the United States, although a substantial share also involved Asian countries. The studies investigated a range of chatbot functions, most commonly generalist assistants ($n = 27$), social ($n = 19$), and transactional ($n = 17$) chatbots, with fewer focusing on health ($n = 10$) or educational ($n = 4$) purposes. In terms of investigated antecedents, chatbot-related factors ($n = 46$) were examined most frequently, followed by user-related ($n = 30$) and message-related ($n = 27$) factors. For detailed descriptions for each study, refer to Appendix D1.

3.4. Antecedents of *human–AI chatbot* relationships: User, chatbot, and message

Tables 3 and 4 summarize predictors of four of the most studied relational outcomes in the reviewed literature. Because the reviewed studies examined a wide range of relational variables, we adopted a

Table 2. Characteristics of included studies.

Category	Subcategory	*n*	Category	Subcategory	*n*	Category	Subcategory	*n*
Study Design	Content analysis	2	Sample size[b]	100 or smaller	25	Function of Chatbots (purpose)	Educational	4
	Experiment (online)	35		100–299	20		Health	10
	Experiment (lab/in-person)	13		300–499	22		Generalist	27
	Experiment (other)	3		500 or larger	6		Social	20
	Survey (online)	24	Sample criteria	Age	23		Transactional	17
	Survey (in-person)	1		Occupation	16	Antecedents examined[c]	Chatbot-related	46
Pub Year	2020 or earlier	14		Location/nationality	12		User-related	30
	2021	10		Experience/ability	36		Message-related	27
	2022	15		Language	7			
	2023	35	Sex/gender	Female: > 60%	15			
	2024	4		Female: 40%–60%	39			
Age[a]	Age mean offered	45		Female: < 40%	16			
	Age mode, range, or % offered	45	Country	European countries	27			
				United States	21			
				Asian countries	18			

Note: $N = 78$.
[a]Studies offering both age mean value and range were duplicated.
[b]Sample sizes reported only for studies where individuals were the unit of analysis.
[c]Studies may examine more than one type of antecedent.

Table 3. Predictors of trust.

Grand category	Subcategory	Specific predictors
Positive associations		
Chatbot	Human-likeliness	Human-Like Empathy[a], Facial Expression (Vs. Body Gesture)[b], Anthropomorphism[ab], Human Representation[ab]
	Presence	Social Presence[b], Authentic Experiences[a]
	Competence	Recommendation(R) Reliability[b], R Accuracy[b], R Choice[a], Cognitive Competency[a], Emotional Competency[a], Service Quality[ab], Task Efficiency[a], Personalization[a], Perceived Intelligence[a], Perceived Expertise[a], Service Attitude[a], Information Migration[ab], Identity Migration[ab]
	General Technology Experiences	Perceived Usefulness[b], Perceived Ease of Use[b], Psychological Safety[a]
	Others	AI Agent (Vs. Other Agents Like Humans)[ab]
Communication		Conversational Styles[b], Empathetic Behaviors[a], Responsiveness[a], Response Quality[b], Communication Quality[b]
User		Propensity To Trust Technology[b]
Relational		Perceived Intimacy[a], Trusting Beliefs[b]
Context		–
Negative associations		
Chatbot	Competence	Fake Information[b], Chatbot Error[b]
	General Technology Experiences	–
Communication		Immediacy of Fake Information Delivery[b]
User		–
Relational		Negative Impression[b]
Context		Task Complexity[b]
Mixed/null associations		
Chatbot	Human-likeliness	Chatbot Sociability[a]
	Competence	Relational Competency[a], Information Seeking[a]
	General Technology Experiences	Perceived Control[ab], Perceived Risk[b]
	Others	Playfulness[a]
Communication		Social Cue[ab]
User		Privacy Concern[a]
Relational		Warmth[a]
Context		Purpose of Chatbots[b]

Note: Numbers indicate references (supplement). Trust is indicated by a variety of variables like cognitive/emotional/relational/benevolence/competence trust, trustworthiness, trusting beliefs, and trusting intentions.
[a]Relational trust.
[b]Instrumental trust. For specific papers, see Appendix D2a.

parsimonious organizing scheme. Specifically, we grouped relational outcomes into two broad categories: relational perceptions and relational behaviors. Relational perceptions include trust and relationship strength, whereas relational behaviors include self-disclosure and aggression. We based this classification on the constructs' conceptual meanings and thematic commonality. While using these

Table 4. Predictors of relationship strength, self-disclosure, and aggression.

Grand category	Subcategory	Specific predictors
Relationship strength		
Positive associations		
Chatbot	Human-Likeliness	AI Sociability, Anthorpormophism
	Presence	Voice, Social Presence
	Competence	AI Supportiveness
	Others	Agent
Message/communication		Chatbot Attachment Behaviors, Interaction Duration, Interaction Intensity
User		User Social Motivation, User Age
Relational		Relationship Quality
Negative associations		
Message/Communication		Frequency Of Questions Asked
Relational		Psychological Distance, Trust
Context		Usage Diversity
Mixed/null		
Chatbot	Human-Likeliness	Naming
User		User Education
		Self-disclosure
Self-disclosure		
Positive associations		
Message/communication		Frequency of Questions Asked (by Chatbot), Depth of Questions Asked (By Chatbot), Active Listening, First-Person Pronoun (Vs. Third-Person Pronoun Or Vs. Control)
User		User (Vs. Nonuser)
Relational		Rapport
Context		Relational Use, Topic (Acquantaincing Vs. Discussing Life Goals)
Mixed/null associations		
Chatbot	Competence	Sensorimotor Competencies, Flexible Action Patterns, Meta-Cognitive Heuristics, Intellectual Competencies
Message/communication		Chatbot Self-Disclosure
Context		Functional Use
Aggression		
Positive associations		
User		User Frustration
Negative associations		
Chatbot	Human-Likeliness	Chatbot Displayed as A Nonhuman User
	Competence	Chatbot Loebner Score
Message/communication		Nonsense Utterance3, Morality in Communication (Persecutor vs. Victim or Helper)
Mixed/null associations		
Chatbot		Humanity, Perceived Humanness
		Gender Known (Vs. Gender Unknown)

Note: Numbers indicate references (supplement). Relationship strength is indicated by a variety of variables like attachment, affinity, dependence, etc. Self-disclosure is indicated by either actual self-disclosure or intentions of self-disclosure. Aggression is indicated by users' aggressive communication behaviors (swear words, sexual comments, etc.). For 16(1), we re-coded the outcomes for parsimonious presentation such that higher communal relationship values indicate higher relationship strength and higher exchange relationship values indicate lower relationship strength. For specific papers, see Appendix D2b.

broad categories for organization, we preserve the original constructs in each study. Tables 3 and 4 report examples of specific variables. This approach is intentionally conservative, summarizing prior work in a way that remains consistent with how the original studies framed their constructs. Therefore, we present these relational outcomes together for parsimonious organization of our findings, and readers should not assume that they are interchangeable relational perceptions or behaviors. For these outcomes, the reviewed studies mostly reported positive predictors, while there were some negative and mixed/null effect predictors. Overall, cues that make chatbots seem more human-like, socially present, and competent tend to show positive associations with relational perceptions (i.e., trust, relationship strength), whereas relational behaviors appear more sensitive to communication structure, message features, and users' momentary states. Notably, the majority of predictors examined were related to chatbot characteristics, with comparatively less attention given to communication, user, relational, or contextual factors. The relational outcomes reviewed here are not exhaustive but include the most common ones.

3.5. Trust

We first review outcomes that are categorized as trust. Trust has heterogeneous definitions and operationalizations, addressing different domains (e.g., general, ability, benevolence; Peng & Wan, 2024) and

distinct evaluative aspects (e.g., affective vs. cognitive; Chen et al., 2025). That said, trust is often defined as a tendency to believe in AI chatbots in fulfilling their roles and meeting user expectations in the studies reviewed.

For chatbot-related predictors, various indicators of human-likeliness (e.g., anthropomorphism; Chen et al., 2025; Li et al., 2023) and social presence in general (e.g., Prakash et al., 2023) were consistently associated with higher trust. Some but not all indicators of competencies (e.g., cognitive, and emotional, Chandra et al., 2022) or general technology experiences (e.g., Prakash et al., 2023) were positively linked to trust outcomes; some were negatively related to trust (e.g., errors; Kim et al., 2023).

For communication-related predictors, quality and supportiveness (empathy: Liao & Yan, 2023; responsiveness; Li et al., 2023) were positively related to higher trust whereas speed of delivering inaccurate information (Amaro et al., 2024) was negatively related to trust. As for user-related predictors, propensity to trust was positively associated with trust (Prakash et al., 2023), but privacy concern was not consistently significantly related to trust (Li et al., 2023). Relational and context factors were also related to trust: higher trusting beliefs (Prakash et al., 2023) and higher intimacy (Qiao et al., 2022) were associated with more trust, and higher negative impression (Kim et al., 2023) and task complexity (Casadei et al., 2022) were associated with lower trust.

Across studies, relational trust is primarily driven by socio-emotional and social connection factors. For example, relational trust tends to increase when chatbots convey empathy, foster psychological safety, feel authentic, and promote relational perceptions such as intimacy. Instrumental trust, on the other hand, depends more on chatbots' task performance and reliability. It increases with usefulness, ease of use, and response and communication quality, and it decreases in the presence of errors and misinformation. Several predictors are associated with both trust types (e.g., anthropomorphism and social cues), suggesting that design features highlighting the chatbot as a social agent can strengthen trust both relationally and instrumentally.

3.6. Relationship strength

Compared with trust, fewer studies examined relationship strength. In this review, relationship strength refers to the emotional closeness of human–AI chatbot relationships, encompassing outcomes such as communal bonding perception (Tschopp et al., 2023) and parasocial relationship perception (Hoffman et al., 2021). Chatbot features related to human-likeliness (e.g., AI sociability; Pentina et al., 2023; Xie et al., 2023), social presence (e.g., voice; Tschopp et al., 2023), competence (AI supportiveness; Leite et al., 2013), were associated with stronger relationships between users and AI chatbots. Communication-wise, longer and more intense interactions (Christoforakos et al., 2021), as well as more display of attachment behaviors (Xie et al., 2023), were associated with stronger relationships. In contrast, having more interactions where chatbots asked extensive questions (Hoffman et al., 2021) was associated with weaker relationship strength. At the user level, older individuals (Tschopp et al., 2023) and those motivated to interact with chatbots socially (Pentina et al., 2023) were more likely to develop stronger relationships with chatbots.

3.7. Self-disclosure

Self-disclosure, as a behavioral relational outcome, refers to the act of revealing personal or private information to AI chatbot (e.g., Croes et al., 2023). Compared with trust, studies on self-disclosure revealed a different pattern of predictors. Chatbot-level predictors, particularly competence-related cues produced mixed findings (e.g., Gieselmann & Sassenberg, 2023), whereas communication-level predictors (e.g., frequency of questions asked by chatbots, first-person pronouns; Croes et al., 2023) were positively related to self-disclosure. Chatbot's self-disclosure, however, was not significantly related to user self-disclosure (Croes et al., 2023). Rapport (Baek & Kim, 2023) and relational use (Xu & Li, 2024) of chatbots were positively related to self-disclosure.

3.8. Aggression

Only a small number of studies examined aggression as a relational outcome. Studies revealed that lower levels of human-likeliness in chatbots were associated with higher aggression (Keijsers et al., 2021). Message features that had higher meaning (Keijsers et al., 2021) and morality (Brendel et al., 2020) were linked to lower aggression. Lastly, more user frustration (Brendel et al., 2023) was associated with higher aggression. For detailed descriptions for each study, refer to Appendix D2a and D2b.

3.9. Human–AI chatbot relationship and social influence outcomes

In the HAIR framework (Figure 1), secondary outcomes refer to socially influenced perceptions and socially influenced behaviors that occur downstream of relational mediators. In Tables 5–7, antecedents (X) map onto user, chatbot, or message characteristics, mediators (M) map onto relational perceptions or behaviors, and secondary outcomes (Y) map onto socially influenced perceptions or behaviors. We summarize each pathway using this structure to make the link to the framework explicit.

3.9.1. Perceptual secondary outcomes

Table 5 summarizes mediation pathways in which relational perceptions link antecedents to socially influenced perceptions. Specifically, Meng and Dai (2021) found that perceived supportiveness of chatbots mediated the effect of emotional support on worry reduction and stress reduction. In a study by Cho et al. (2022), perceptions of active listening by a chatbot mediated the effect of backchanneling on users' enthusiastic emotional states. Furthermore, this study explored serial mediation effects and found that active listening backchanneling influenced perceived usability through perceived active listening and perceived emotional support but not through the serial mediation of perceived active listening and positive emotional words. Similarly, Park, Lee, and Chung (2023) found that social rapport perceptions mediated the effect of chatbot anthropomorphism on both counseling satisfaction and reuse intention.

3.9.2. Behavioral secondary outcomes

Tables 6 and 7 summarize mediation pathways in which relational perceptions link antecedents to socially influenced behaviors. Trust was the most frequently examined mediator. Trust was most frequently studied relational mediator. Prakash et al. (2023) reported that trusting beliefs mediated the effects of perceived usefulness, perceived ease of use, social presence, and propensity to trust technology on users' intention to continue interacting with chatbots. However, trust did not mediate the effect of privacy risk on continuance intention. Similarly, Qiao et al. (2022) demonstrated that trust mediated the effects of perceived intimacy on both the intention to utilize the chatbot and the intention to explore. Baek and Kim (2023) found that trust mediated the effects of information-seeking and task efficacy on continuance intention, but it did not mediate the effects of personalization, social interaction, or playfulness.

Table 5. Mediation effects via human–AI relationship: Perceptual outcomes.

Mediator (M)	Antecedent (X)	Secondary outcomes: perceptual (Y)	Results		Article
			Sig	Effect	
Active listening	Backchanneling	Enthusiasm	√	Serial indirect effect = 0.42, 95% CI [0.0046, 1.0053]	Cho et al. (2022)
Active listening – Perceived emotional support (serial)		Perceived usability	√	Serial indirect effect: 0.35, 95% CI [0.0149, 0.9645]	Cho et al. (2022)
Active listening – Positive emotional words (serial)		Perceived usability	√	Serial indirect effect: 0.02, 95% CI [−0.1693, 0.3737]	Cho et al. (2022)
Perceived Supportiveness	Emotional support	Stress reduction	√	B = .12, SE = .05, 95% CI [.04, .23]	Meng and Dai (2021)
		Worry reduction	√	B = .21, SE = .08, 95% CI [.07, .37]	Meng and Dai (2021)
Social rapport	Anthropomorphism	Counseling satisfaction	√	0.26, SE = 0.07, 95% CI [0.12, 0.41]	Prakash et al. (2023)

Notes: √: studies where the indirect effect was reported with $p < 0.05$.

Table 6. Mediation effects via human-AI relationship: Behavioral outcomes (trust as the mediator).

			Results				
					Direct effects		
Mediator (M)	Antecedent (X)	Secondary outcomes: behavioral (Y)	Sig	Indirect effects	X → M	M →Y	Article
Trust (Trusting belief)	Perceived usefulness	Continuance intention	√	β = 0.116, t = 3.618	NA	NA	Prakash et al. (2023)
	Perceived ease of use		√	β = 0.144, t = 4.304	NA	NA	Prakash et al. (2023)
	Social presence		√	β = 0.065, t = 2.529	NA	NA	Prakash et al. (2023)
	Propensity to trust technology		√	β = 0.045, t = 1.951	NA	NA	Prakash et al. (2023)
Perceived trust worthiness → Satisfaction with recommendation (serial)	Salience of ChatGPT: positive vs negative ethical issue	Visit intention to recommended destination	√	95% CI [−0.351, −0.009]	NA	NA	Kim et al. (2023)
	Salience of ChatGPT: positive and quality issue		√	95% CI [−0.465, −0.109]	NA	NA	Kim et al. (2023)
Trust	Perceived intimacy	Intention to utilize	√	NA	β = 0.47, $p < 0.001$	β = 0.29, $p < 0.001$	Qiao et al. (2022)
	Perceived intimacy	Intention to explore	√	NA	β = 0.47, $p < 0.001$	β = 0.20, $p < 0.01$	Qiao et al. (2022)
	Information seeking	Continuance intention	√	NA	Standardized estimate = 0.291, critical ratio = 1.945, $p<0.10$	Standardized estimate = 0.187, critical ratio = 4.195, $p < 0.01$	Baek and Kim (2023)
	Task efficiency	Continuance intention	√	NA	Standardized estimate = 0.522, critical ratio = 3.073, $p <0.01$	Standardized estimate = 0.187, critical ratio = 4.195, $p < 0.01$	Baek and Kim (2023)
	Human representation	Intention to comply with chatbot-recommendations	√	B = 0.09, SE = 0.05, 95% CI [0.01, 0.19]	NA	NA	Park et al. (2022)
Trusting belief	Privacy risk	Continuance intention	×	Not statistically significant	NA	NA	Prakash et al. (2023)
Trust	Personalization	Continuance intention	×	NA	Standardized estimate = −0.021, critical ratio = −0.029, $p>0.10$	Standardized estimate = 0.187, critical ratio = 4.195, $p < 0.01$	Baek and Kim (2023)
	Social interaction		×	NA	Standardized estimate = 0.237, critical ratio = 1.107, $p>0.10$	Standardized estimate = 0.187, critical ratio = 4.195, $p < 0.01$	Baek and Kim (2023)
	Playfulness		×	NA	Standardized estimate = 0.041, critical ratio = 0.083, $p>0.10$)	Standardized estimate = 0.187, critical ratio = 4.195, $p < 0.01$	Baek and Kim (2023)
	Perceived intelligence of AI	Adoption intention of SRH chatbots	×	NA	β = 0.583; t = 13.115; $p < 0.001$	β = −0.018, t = 0.236, $p = 0.407$	Liew et al. (2023)
	Anthropomorphism		×	NA	β = 0.036; t = 0.692; $p = 0.245$	β = −0.018, t = 0.236, $p = 0.407$	Liew et al. (2023)

Notes: √: studies where the indirect effect was reported with $p < 0.05$, or where the main effects of X on M and M on Y were both significant ($p < 0.05$); ×: studies where the indirect effect was reported with $p > 0.05$, or where either the effect of X on M or M on Y was not significant ($p > 0.05$).

Table 7. Mediation effects via human–AI relationship: Behavioral outcomes (other relational mediators).

Mediator (M)	Antecedent (X)	Secondary outcomes: Behavioral (Y)	Results				
			Sig	Indirect effects	Direct effects: X → M	Direct effects: M →Y	Article
Psychological distance	Human representation	Intention to comply with health recommendations	√	B = 0.07, SE = 0.04, 95% CI [0.02, 0.16]	NA	NA	Park et al. (2022)
Social attraction	Emotional support	Purchase intention	√	NA	$\beta = 0.44$, $p < 0.01$	$\beta = 0.34$, $p < 0.01$	Lee et al. (2022)
	Informational support		√	NA	$\beta = 0.23$, $p < 0.01$	$\beta = 0.34$, $p < 0.01$	Lee et al. (2022)
	Esteem support		√	NA	$\beta = 0.33$, $p < 0.01$	$\beta = 0.34$, $p < 0.01$	Lee et al. (2022)
Emotional credibility	Emotional support		√	NA	$\beta = 0.36$, $p < 0.01$	$\beta = 0.24$, $p < 0.05$	Lee et al. (2022)
	Informational support		√	NA	$\beta = 0.26$, $p < 0.01$	$\beta = 0.24$, $p < 0.05$	Lee et al. (2022)
	Esteem support		√	NA	$\beta = 0.30$, $p < 0.01$	$\beta = 0.24$, $p < 0.05$	Lee et al. (2022)
Perceived knowledge	Disclosure timing	Call length	√	NA	$B = -2.837$, $p < 0.001$	$B = 2.587$, $p < 0.01$	Luo et al. (2019)
Perceived empathy			√	NA	$B = -2.371$, $p < 0.001$	$B = 1.822$, $p < 0.001$	Luo et al. (2019)
Social rapport	Anthropomorphism	Reuse intention	√	B = 0.26, SE = 0.08, 95% CI [0.12, 0.42]	NA	NA	Park et al. (2023)

Notes: √: studies where the indirect effect was reported with $p < 0.05$, or where the main effects of X on M and M on Y were both significant ($p < 0.05$).

Other studies reported indirect effects via trust or related relational mediators for outcomes. Kim et al. (2023) examined the serial mediation effects of perceived trustworthiness and satisfaction with recommendations in relation to salience of ChatGPT issues. They found statistically significant serial mediation effects for the relationship between positive vs. negative ethical issues and users' intention to visit recommended destinations, as well as between positive vs. negative quality issues and visit intention. Similarly, Park et al. (2022) found that trust mediated the relationship between human-like representation (vs. machine-like) and intention to comply with chatbot recommendations. Additionally, a significant serial mediation effect was observed, where trust and psychological distance mediated the relationship between human-like representation and intention to comply. However, Liew et al. (2023) found that trust did not mediate the effects of perceived intelligence of AI or anthropomorphism on users' intention to adopt sexual and reproductive health information chatbots.

Beyond trust, additional relational mediators were identified, including psychological distance, social attraction, emotional credibility, perceived knowledge, and perceived empathy, linking antecedents to behavioral outcomes such as compliance intention, purchase intention, and call length (Table 7). Park and colleagues (2022) found that psychological distance mediated the relationship between human-like representation and users' intention to comply with health recommendations. In a study by Lee et al. (2022), social attraction mediated the associations between emotional support, informational support, and esteem support with purchase intention. Additionally, emotional credibility was identified as a mediator for these associations. Finally, Luo et al. (2019) demonstrated that perceived knowledge and perceived empathy mediated the relationship between the timing of disclosing the chatbot's identity as a conversational partner and the length of the call.

4. Discussion

As AI chatbots are evolving rapidly and research on their potential for social influence expands, having higher-level theoretical frameworks becomes increasingly salient. Despite the surge in studies, many focus on isolated aspects of chatbot interaction. To address this, we draw upon Bickmore and Picard's notion of relational agents (2005) and forward a relationship-centered framework, the HAIR framework of social influence. This relationship-centered framework guided our systematic review of studies examining the development of human–AI chatbot relationships. Our findings underscore the multifaceted nature of human–AI chatbot relationships, with studies spanning diverse contexts and approaches. Notably, message-level predictors remain underexplored, despite their potential to yield practical insights for chatbot designs. This synthesis revealed that trust and perceived social support frequently function as relational mediators through which chatbots exert social influence. Among these, trust stood out as the most prominent mediator, consistently linked to persuasion outcomes and intention of continue using the chatbots.

4.1. A *human–AI chatbot* relationships perspective to social influence

Our review distinguishes itself from prior systematic reviews that examined human–AI chatbot relationships by treating relational perceptions and behaviors not merely as outcomes (e.g., Chaturvedi et al., 2023; Ng & Zhang, 2025) but as *pathways* through which chatbots exert social influence. We identified three primary purposes of chatbot deployment: to gain compliance, provide social support and promote mental health, and encourage continued chatbot use.

Across these outcomes, relationship factors such as trust and perceived social support were generally found to mediate the effects of antecedents, including user, chatbot, and message characteristics. While scholars often compare human–AI chatbot interactions to interpersonal communication, our synthesis suggests a more nuanced pattern. Human-likeness or anthropomorphism consistently predicted trust, yet findings were more mixed for other relational outcomes such as relationship strength, self-disclosure, and aggression. This pattern suggests that human-likeness may be a necessary but not sufficient condition for developing strong human–AI relationships. Therefore, developing strong relationships with chatbots should be a central consideration when researchers or designers seek to implement chatbots for specific purposes. Indeed, the influence of relational factors is not unique to human–AI

interactions. Relationship dynamics have long been recognized in social influence research as critical drivers of persuasion and behavior change (e.g., Cialdini & Goldstein, 2004; Latané, 1981).

4.2. Influence of user characteristics

Our results showed that user characteristics can play a meaningful role in shaping human–AI chatbot relationships. Consistent with broader human-computer interaction research, individual differences such as propensity to trust emerged as important facilitators of relational outcomes (Prakash et al., 2023). For instance, users who generally report higher tendencies to trust technology were more likely to extend trust toward AI chatbots, suggesting that these baseline user inclinations toward technologies may transfer into human–AI chatbot contexts. User demographics and motivations also shaped relational outcomes. Age was positively related to relationship strength, with older users more likely to form close, enduring bonds with chatbots. This aligns with prior findings that older adults may value companionship and supportive communication in digital interactions, thereby enhancing relational depth (Tschopp et al., 2023). Similarly, individuals who reported higher social motivations for interacting with chatbots developed stronger relationships, underscoring the importance of understanding user goals when assessing relational dynamics.

These findings collectively suggest that human–AI relationships cannot be fully explained by chatbot design alone. Instead, users' dispositional traits, demographic factors, and motivational orientations moderate whether relational communication translates into trust, self-disclosure, or deeper relationship strength. Future research should further investigate the intersection of these characteristics with chatbot and communication features, particularly through longitudinal and cross-cultural designs that can capture the variability and evolution of these relationships across diverse user populations.

4.3. Influence of chatbot's characteristics

Across all functional categories of chatbots, including educational, social, transactional, health, and generalist, research consistently prioritizes chatbot-related antecedents. This reflects a prevailing assumption in the field: that a chatbot's design and core features are the primary determinants of successful interaction. For educational chatbots, this often translates into an emphasis on pedagogical effectiveness; for transactional chatbots, efficiency and accuracy are central; for health chatbots, trustworthiness and personalized support are critical; and for social chatbots, a believable personality and simulated empathy are important. Even for generalist chatbots, adaptability is itself a design choice.

This strong focus on chatbot design suggests several implications. First, it highlights the dominance of a technology-centered perspective, where improvements to functionality are expected to drive relational and behavioral outcomes. Second, it raises the possibility that other antecedents, such as user dispositions or communication dynamics, may be undervalued in current research, potentially limiting the field's understanding of human–AI relationship building. Third, the emphasis on design reflects a tendency toward short-term, experimental evaluations of isolated features rather than investigations of how design interacts with user characteristics and relational processes over time. Finally, this pattern underscores the need for integrative approaches, where design is studied not in isolation but as part of a broader system in which user characteristics and relational communication are equally central.

4.4. A relational communication perspective to *human–AI chatbot* relationships

As a central aim of this review, we find that relational mediators, especially trust, generally translated the effects of chatbot characteristics (e.g., anthropomorphism: Park et al., 2022), user characteristics (e.g., propensity to trust technology: Prakash et al., 2023), and message characteristics (e.g., emotional support: Meng & Dai, 2021) to secondary perceptual and behavioral outcomes. This pattern highlights the value of incorporating relational processes when designing and evaluating chatbots for social influence. These findings also echo prior work on relationship dynamics in social influence, which positions the relationship as a critical driver of persuasion and behavior change (e.g., Cialdini & Goldstein, 2004; Latané, 1981).

This review further expands upon previous systematic reviews by reviewing message-based determinants of human–AI chatbot relationship development. Although relatively few studies have examined the effects of communication content, those that did consistently found that empathic and responsive chatbot messages foster stronger bonds between humans and AI (e.g., Cho et al., 2022; Li et al., 2023; Liao & Yan, 2023), which in turn are linked to various influence outcomes. These findings underscore that, beyond chatbot characteristics and users' dispositions, it is important to pay more attention to communicative exchanges between humans and chatbot. Many constructs that appear to be chatbot-level attributes are operationalized in prior work as features of conversational turns or interaction sequences, such as responsiveness, empathic behaviors, and personalization enacted in replies. We therefore treat these as message characteristics when they are measured or manipulated at the level of interaction, and reserve chatbot characteristics for stable system or interface properties.

At the same time, because relationships play central roles in AI chatbot social influence, it may be promising to apply interpersonal communication theories to human–AI chatbot relationships, while carefully testing where these theories translate and where they require revision. For example, whether those of supportive communication that underline importance of responsiveness (Holmstrom & Lim, 2023), can be readily applied to human–AI chatbot communication contexts or not. While early research on human-machine communication held that humans tend to automatically apply human-human social scripts to interactions with machines (Reeves & Nass, 1996), more recent work argues that as people accumulate experience with machines, the scripts guiding human-AI interaction may have evolved to be different from those guiding human–AI chatbot interaction (Gambino et al., 2020). If so, users may hold distinct expectations for empathic and responsive chatbot communication compared to interactions with fellow humans. Exploring these social interaction scripts' differences represents a promising direction for future research.

4.5. Implications

This review contributes to future research in several ways. First, it offers scholars a systematic guide to studying how relational processes can be harnessed for social influence. Rather than treating relational constructs as incidental outcomes, our framework, *the HAIR Framework of Social Influence,* positions them as central pathways through which chatbots build personal relationships with users and exert influence. Future research can build on this framework to design studies that examine how different antecedents including user, chatbot, and message characteristics shape relational dynamics and, in turn, behavioral outcomes.

For chatbot developers, the proposed framework offers design principles for creating chatbots that serve relational functions across contexts. Beyond technical performance, developers should embed features that foster trust, rapport, and responsiveness, with future work exploring how message strategies such as self-disclosure, personalization, and empathy cues can be operationalized ethically. This distinction implies two complementary design levels. System- level choices should target stable relational affordances, such as role framing, identity disclosure, modality, and reliability and error handling features, whereas message-level design should specify how empathy, active listening, contingency, calibrated self-disclosure, and personalization are enacted in the content of responses. For service providers in sectors like health and education, our findings highlight that effective outcomes, whether compliance, support, or sustained engagement, depend on cultivating strong human–chatbot relationships. This relational leverage also carries risk. Trust and perceived social support may heighten overreliance when chatbot interaction becomes a primary source of emotional reassurance or guidance. Work on social companion chatbots documents emotional dependence and related harms (Hu et al., 2023), indicating that relational benefits and relational risks can coexist (Laestadius et al., 2024). Because most existing studies are short term, future research should test relationship centered interventions in real world settings using longitudinal designs and outcome measures that track reliance trajectories, potential displacement of offline support, and the boundary between supportive engagement and problematic dependence.

4.6. Limitations and future directions

The current review has several limitations that should be noted. First, we restricted our search to peer-reviewed, full-text studies published in English and did not include gray literature, which may limit the comprehensiveness of the evidence base. A key assumption of many studies is that AI chatbots are used to optimize individuals' hedonic experiences. With this being natural in individualistic cultures, we wonder whether AI chatbots may also serve collective purposes (e.g., enhancing community solidarity) in collectivist cultures; however, the second aspect is less studied and may be enhanced in future research. Second, due to the heterogeneity of outcome measures and the limited number of randomized controlled trials (RCTs), we were unable to conduct a meta-analysis, which constrains our ability to draw firm conclusions about the effects of AI chatbots on human–AI chatbot relationships. We encourage future research to focus on specific relational outcomes, such as trust and self-disclosure, and to conduct meta-analyses to quantify the magnitude and boundary conditions of these effects as the evidence base continues to grow. Third, the risk of bias assessment identified several recurring methodological concerns. Many randomized trials did not clearly report baseline comparability across groups, and a large proportion of descriptive studies relied on samples with limited representativeness. Because our findings were not weighted or restricted by study quality, conclusions reflect the full range of methodological rigor and should therefore be interpreted cautiously. Future work should prioritize stronger sampling procedures and clearer reporting practices. In addition, many of the included studies examined interactions of very short duration, making it difficult to draw conclusions about the development of human–AI chatbot relationships. Because relationships are inherently dynamic and tend to evolve over time, short-term designs limit our ability to assess whether initial relational perceptions translate into sustained relationships. Future work should employ longer-term study designs to better capture the trajectories of relationship building with chatbots.

This review also highlights key limitations in the current state of research. A primary concern is the inconsistency, and sometimes absence, of clear conceptual and operational definitions for variables, including antecedents, mediators, and outcomes. These inconsistencies pose challenges to synthesizing prior research and accumulating coherent findings. For example, trust can be defined and operationalized both generally and multi-dimensionally (Rapp et al., 2021), with different implications depending on the specific definitions and operationalizations used. As Rapp and his colleagues pointed out, research has often defined and measured trust in a specific way without sufficient justifications of the choice, thus making undue generalizations of their findings. In addition, chatbots have been seen both as service providers or tools (e.g., Adam et al., 2021) and relational agents (Brandtzaeg et al., 2022). Trust in service providers can be more instrumental, whereas trust in social agents can be more interpersonal and relational. Failure to distinguish between instrumental and relational trust can lead to inaccurate generalizations in research syntheses (Rapp et al., 2021). Future research should prioritize theoretical clarity by explicitly distinguishing between different types of trust (e.g., instrumental vs. relational) and aligning measurement with these definitions. Doing so will allow findings to be meaningfully compared across studies and accumulated into broader theoretical insights.

5. Conclusion

This systematic review synthesizes research on human–AI chatbot relationships, identifying key antecedents, relational mediators, and social influence outcomes. It highlights how relationships with chatbots serve as a central process through which social influence unfolds, with message characteristics playing an especially understudied role. In doing so, the review informs both theoretical advancement and practical design of AI chatbots across diverse contexts.

Future research on human–AI chatbot relationships should explicitly integrate interpersonal communication theory with human-computer interaction research. Interpersonal communication perspectives identify message mechanisms that produce relational states, whereas HCI specifies the design and system constraints that determine whether these cues support sustained, appropriate use. We outline three integration pathways. First, construct alignment is needed to enable cumulative evidence, including clear distinctions between relational trust and instrumental trust and between perceived support and

usability or satisfaction. Second, boundary conditions should be theorized and tested, including how errors, transparency cues, privacy costs, and prior AI experience shape the transfer of human–human interaction scripts. Third, methodological integration should link message-level analysis with longitudinal, ecologically valid evaluations that capture relationship trajectories and downstream influence. This agenda positions the HAIR framework as a cumulative scaffold for explaining when relational communication promotes beneficial influence versus when it produces miscalibration and overreliance.

Ethical approval

This study is a systematic review of previously published research. No new data involving human participants were collected or analyzed. Therefore, institutional ethical approval was not required.

Author contributions

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Data availability statement

Data is available upon reasonable request from the correspondence author.

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